09.26

EXP 10 (a): INTERNETWORKING WITH ROUTERS

AIM:

Internetworking with routers in CISCO PACKET TRACER simulator.

PROCEDURE:

Step 1: Configuring Router

1. Open cisco packet tracer and place one router and 2 PCs.
2. Select the router and open the CLI, and press enter to access the command prompt.
3. Enter privileged mode using enable.
4. Enter global configuration mode using configure terminal.
5. Configure FastEthernet 0/0 with:
   - IP address : 192.168.10.1
   - Subnet Mask : 255.255.255.0
6. Use no shutdown to activate the interface.
7. Configure FastEthernet 0/1 with:
   - IP address : 192.168.20.1
   - Subnet Mask : 255.255.255.0
8. Use no shutdown to activate the interface.

Step 2 : Configuring PC's

Device	IP Address	Subnet Mask	Default Gateway
PC_0	192.168.10.2	255.255.255.0	192.168.10.1
PC_1	192.168.20.2	255.255.255.0	192.168.20.1

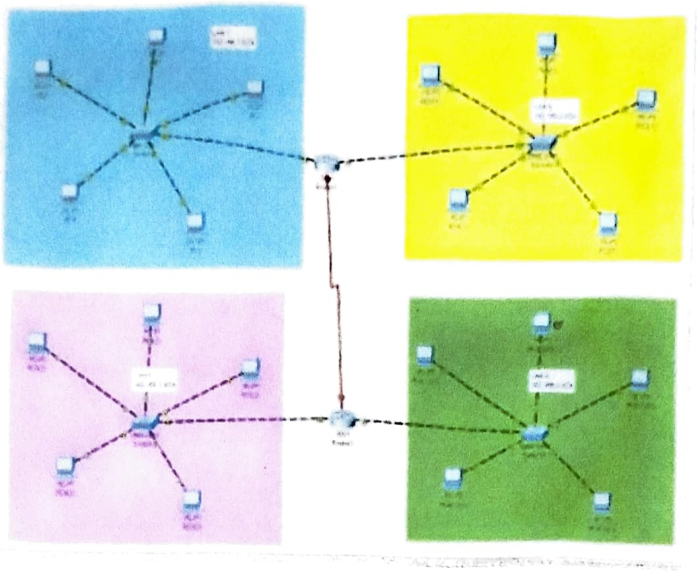
Step 3 : Connecting PC_0 with Router

1. Connect PC_0 FastEthernet to Router1 FastEthernet 0/0 using a copper straight - through cable.
2. Connect PC_1 FastEthernet to Router 1 FastEthernet 0/1 using a copper straight - through cable.
3. Wait until the interfaces become active.
4. Use simulation mode and send a style PDU from PC_0 to PC_1.
5. Verify that the PDU reaches PC_1 and an acknowledgement is received back to PC0.

Configuration Table:

Router Configuration

Device	FastEthernet 0/0	Subnet Mask	FastEthernet 0/1	Subnet Mask
Router 1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14

RESULT:

Thus, a simple internetwork was successfully designed and configured using a router in CISCO PACKET TRACER.

8.09.26

EXP 10(b): INTERNETWORK USING WIRELESS ROUTER

AIM:

To design and configure an internetwork using a wireless router, DHCP server and internet cloud in Cisco Packet Tracer.

PROCEDURE:

Step 1: Build the network topology

1. Open cisco packet tracer and add the devices like PC, Router, laptop, cable modem, internet cloud and server.
2. Connect the devices using the appropriate cables.

Step 2: Configure the router

1. Wireless Router → GUI → wireless
   Network name → Homenetwork.
2. Go to setup tap and ensure DHCP is enabled.
3. Configure the server as 208.67.220.220
4. Show the settings.

Step 3: Configure the laptop

1. Laptop → Physical tab → poweroff → remove ethernet cable and add WPC300N wireless module → power on the laptop.
2. Desktop → PC wireless. select connect tab and select homenetwork.

Step 4: Configure the PC

1. PC → Desktop → IP configuration. Select DHCP.
2. The PC automatically receives an IP address from the wireless router.
3. Use the command prompt and ipconfig /all to verify the configuration.

Step 5: Configure the Internet Cloud.

1. Open the internet cloud.
2. Config → cable
3. Set :- From post :- Coaxial To post : Ethernet
4. Click Add
5. Go to Ethernet and select cable as Provider network.

Step 6: Configure the cisco.com server

DHCP configuration:

Parameter	Value
Pool Name	DHCP Pool
Default Gateway	208.67.220.220
DNS Server	208.67.220.220
Starting IP	208.67.220.1
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add

DNS configuration:

Parameter	Value
Name	cisco.com
Type	A Record
Address	208.67.220.220

Click Add.

Step 7: Verify Connectivity

RESULT:

Thus, the wireless router, DHCP server and Internet cloud were successfully configured in cisco packet tracer and connectivity was verified successfully.